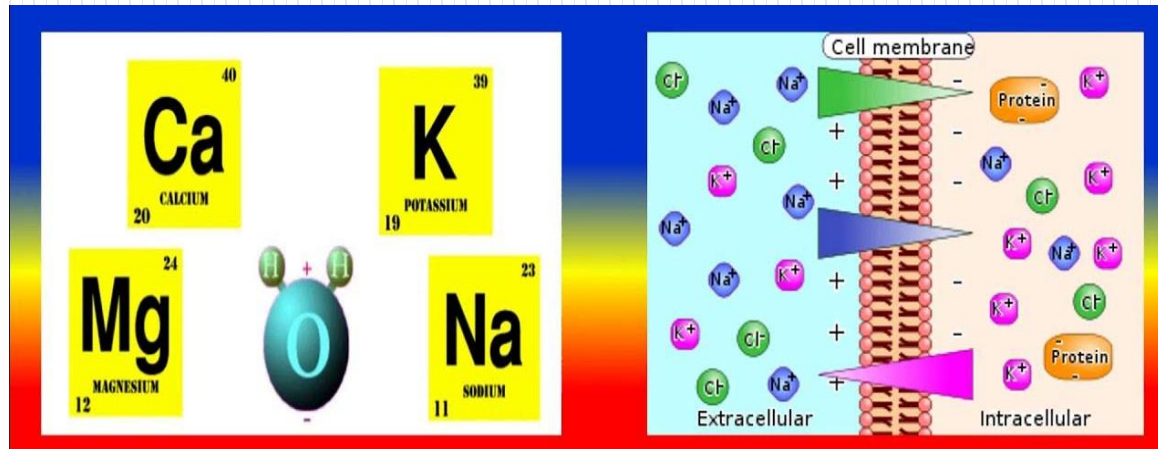


Management of patient with fluid & electrolytes imbalance



Dr . Bardes Shaban



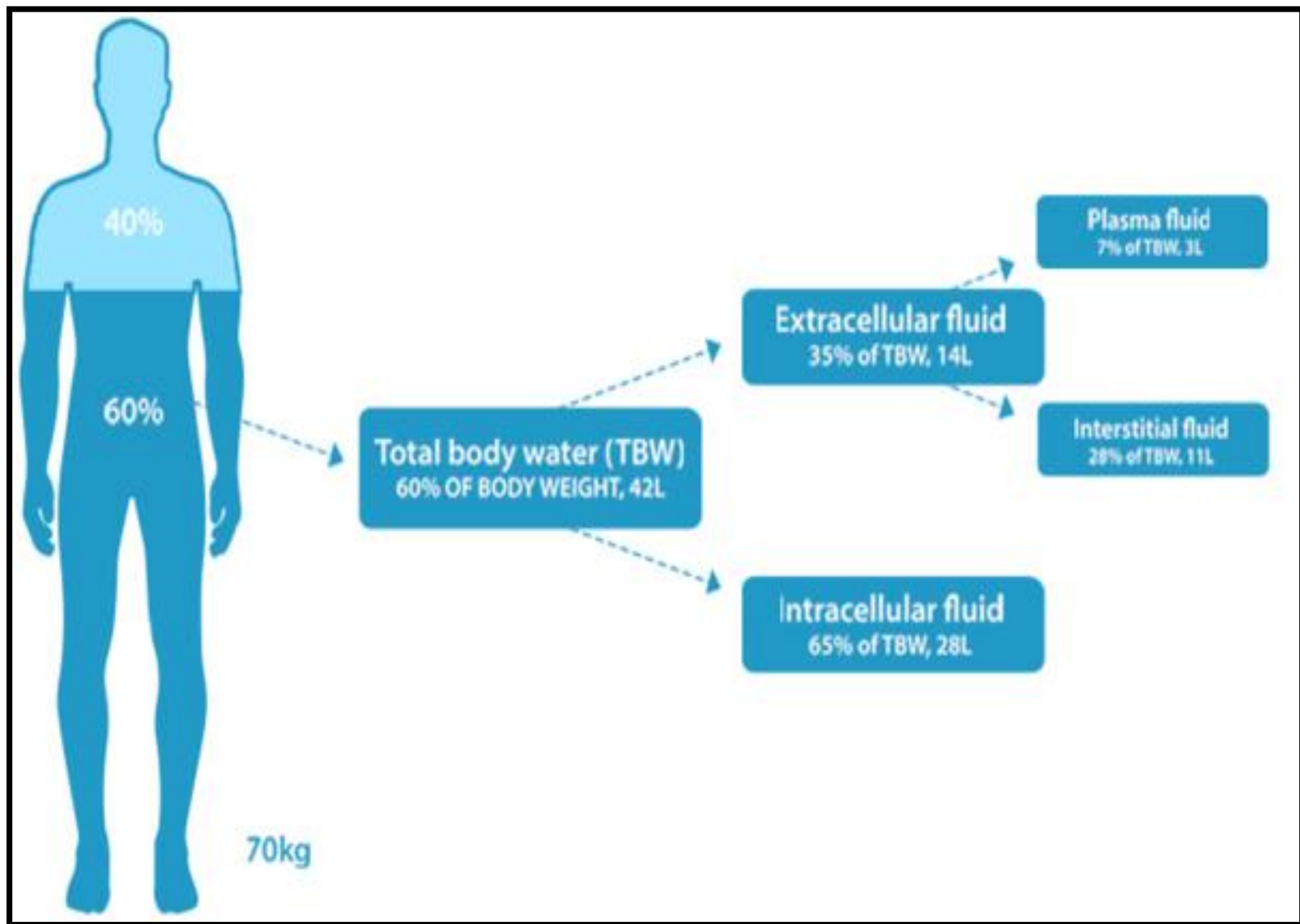
الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف

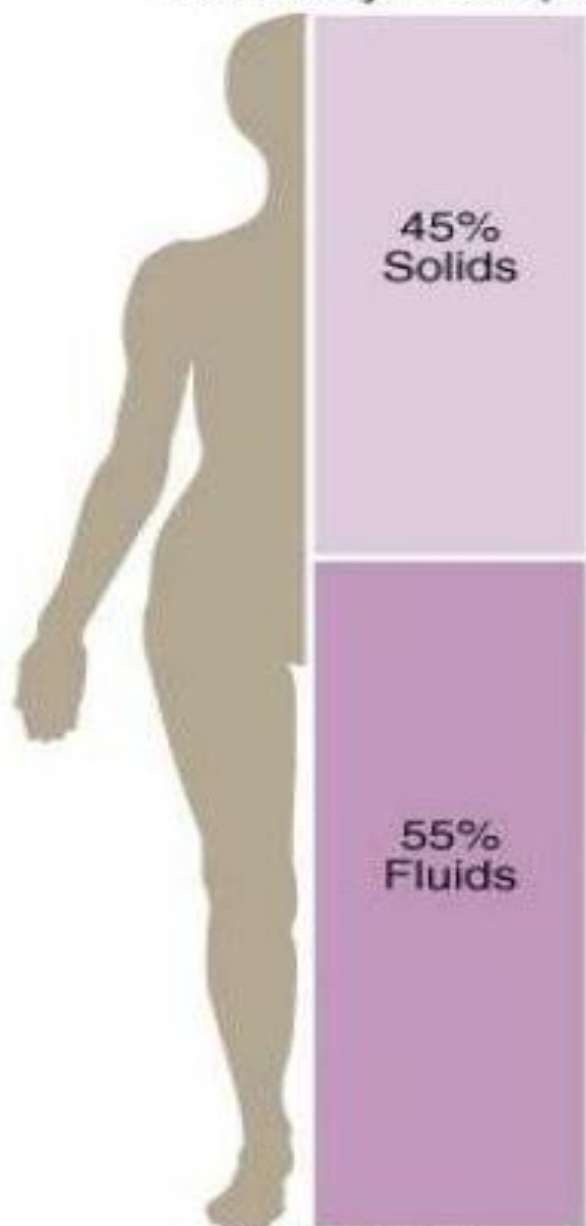
OUTLINES

- ◉ Distribution of body water.
- ◉ Movement of fluids and electrolytes in the body.
- ◉ Fluid imbalances
- ◉ Electrolyte Imbalances

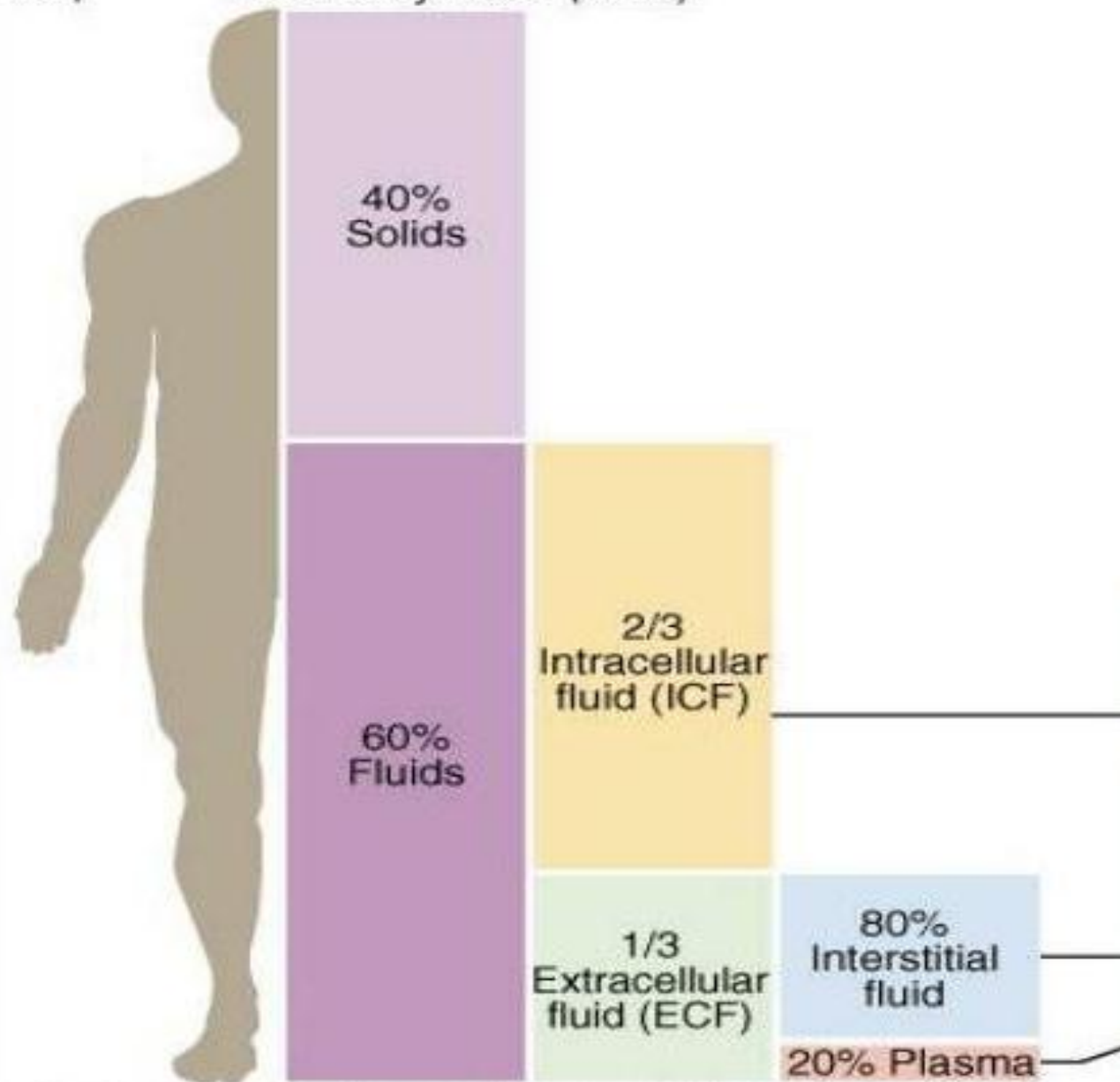


Distribution of body water

Total body mass (female)

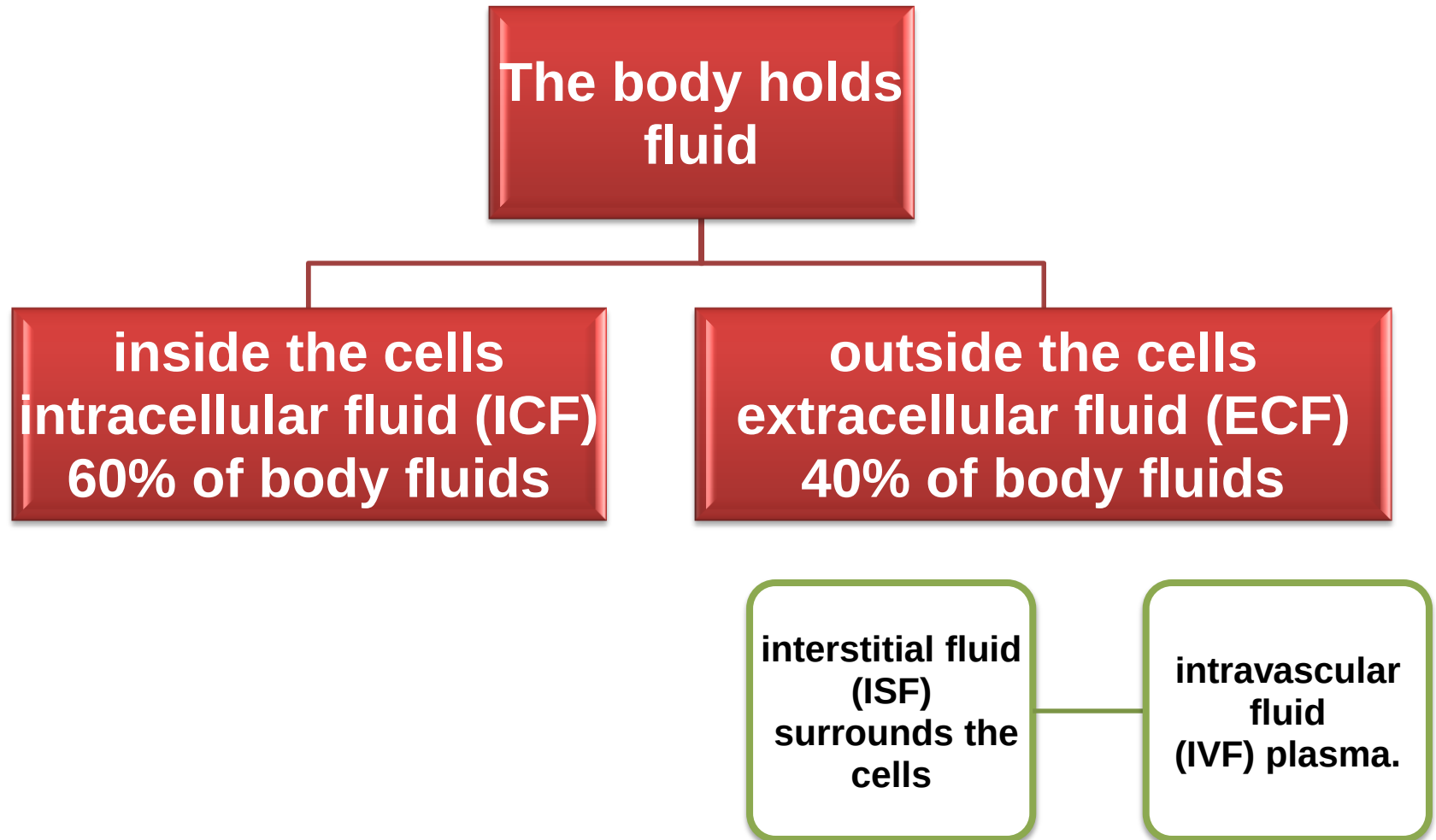


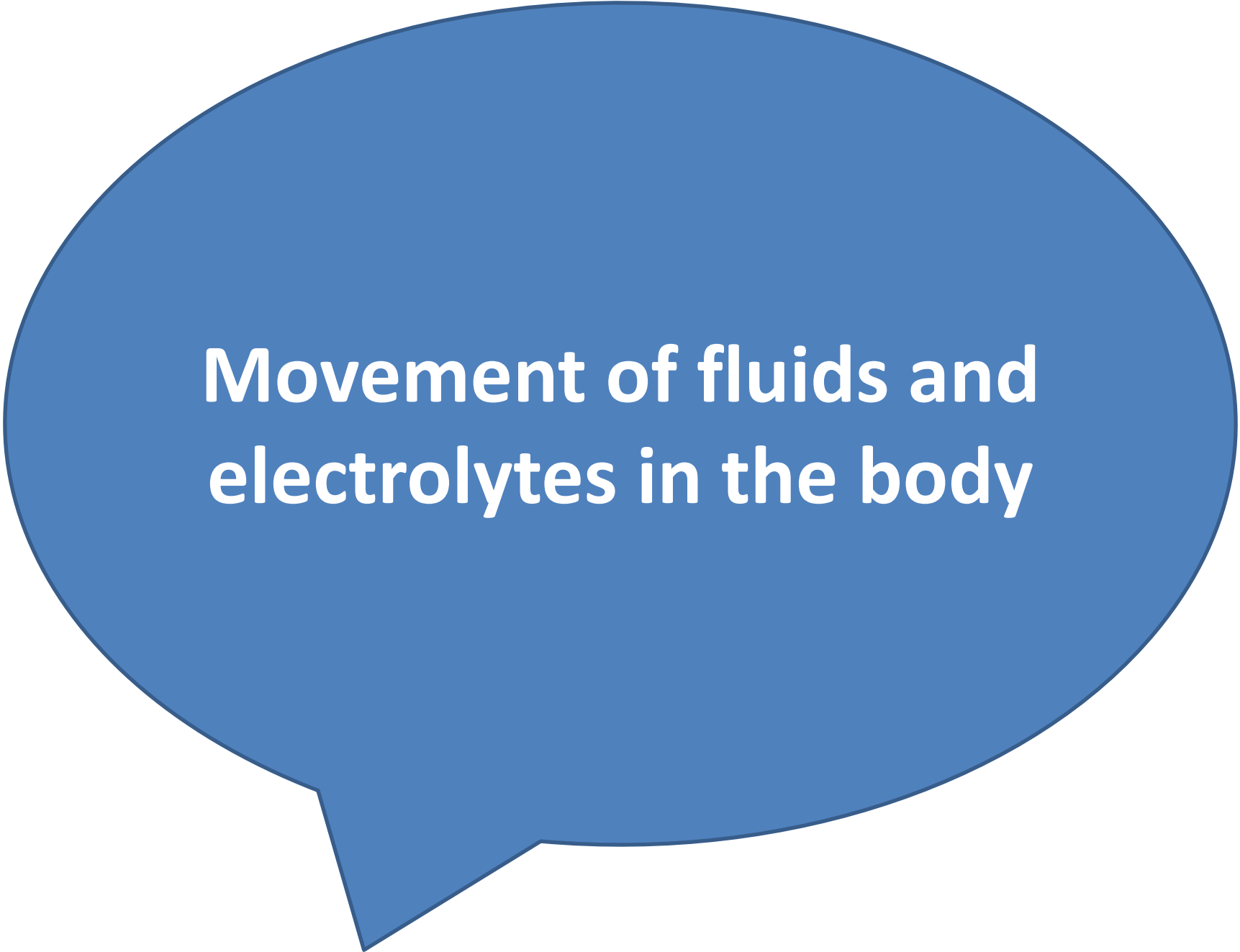
Total body mass (male)



(a) Distribution of body water in an average lean, adult female and male

Body fluids compartment





**Movement of fluids and
electrolytes in the body**

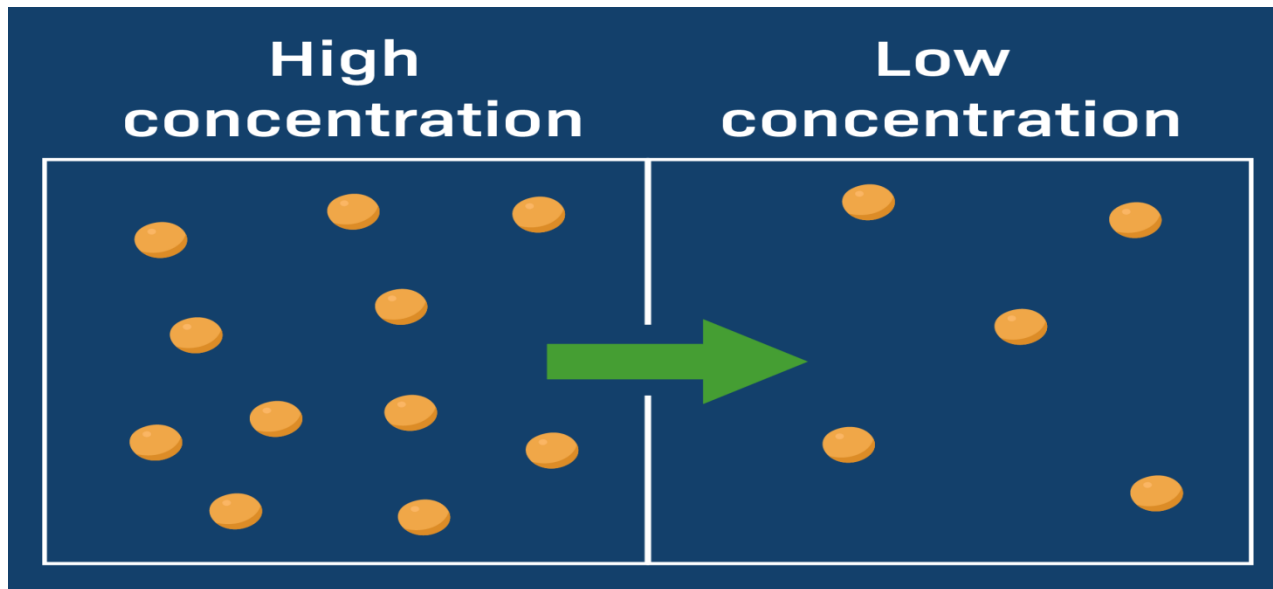
Osmosis

The movement of fluid across cell membrane from the **less concentrated** solution to the **more concentrated** solution



Diffusion

Movement of a solutes (dissolved particles)
across cell membrane from **higher concentration**
to **less concentration**



Fluid Types

```
graph TD; A[Fluid Types] --> B[Isotonic]; A --> C[Hypotonic]; A --> D[Hypertonic]; B --> E["A solution that has about the same concentration of particles as plasma"]; C --> F["A solution has less osmolality or concentration than plasma."]; D --> G["A solution has a greater osmolality or concentration than plasma."];
```

Isotonic

A solution that has about **the same** concentration of particles as plasma

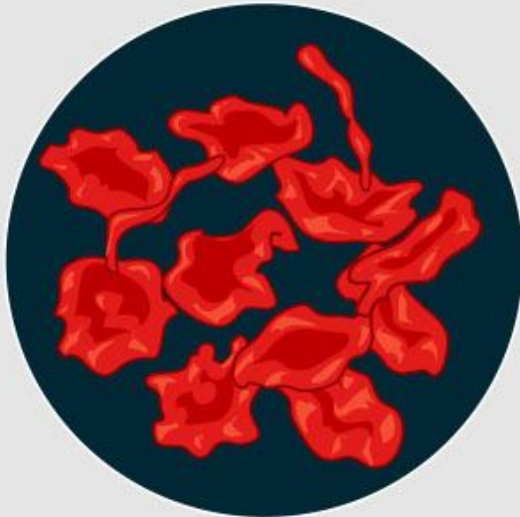
Hypotonic

A solution has **less** osmolality or concentration than plasma.

Hypertonic

A solution has a **greater** osmolality or concentration than plasma.

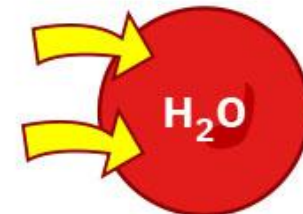
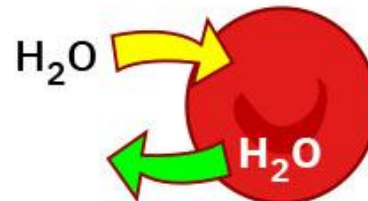
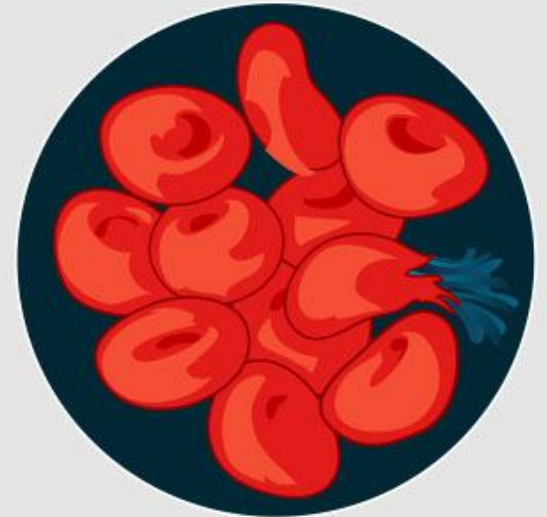
Hypertonic



Isotonic



Hypotonic



Fluid Imbalances

Fluid volume deficits (FVD)

Hypovolemia

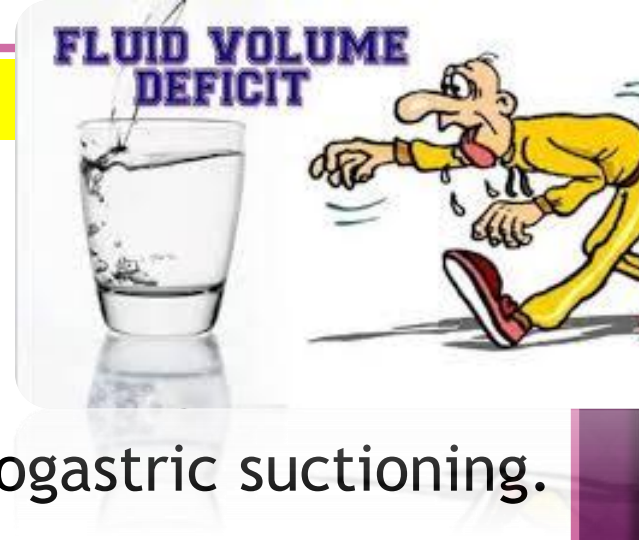
Dehydration

Fluid volume excess (FVE)

Hypervolemia

Overhydration

Fluid volume deficit



Fluid output exceeds the fluid intake

Causes:

- **Gastrointestinal losses:** vomiting, nasogastric suctioning.
Diarrhea.
- **Skin losses:** diaphoresis.
- **Renal losses:** diuretic therapy, diabetes insipidus.
- **Hemorrhage.**
- **Altered intake:** such as nothing by mouth (NPO).
- **Frequent enemas**

Fluid volume deficit

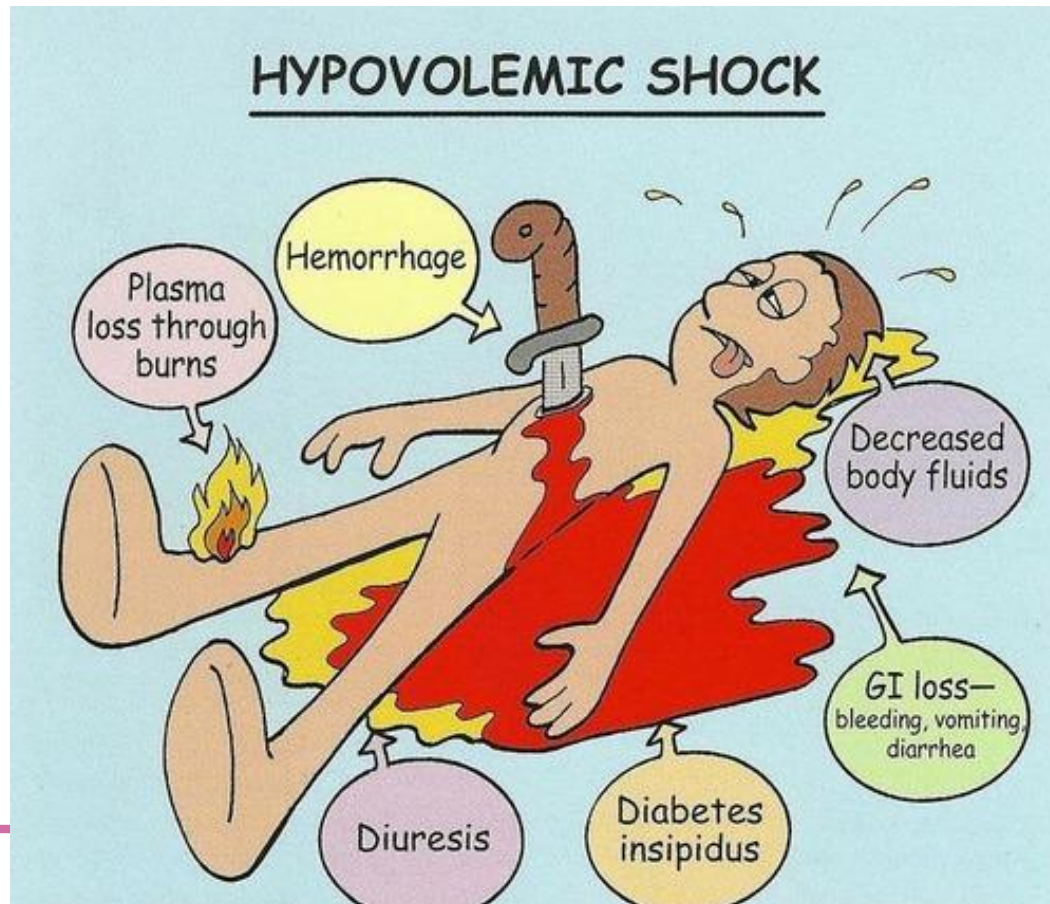
Clinical manifestation:

- **Vital signs:** hyperthermia, thready pulse, hypotension, tachypnea, and tachycardia
- **Neuromusculoskeletal:** Dizziness, syncope, confusion, weakness, fatigue
- **Gastrointestinal:** Thirst, dry tongue, nausea, vomiting, anorexia, weight loss, and constipation
- **Renal:** Oliguria
- Poor skin turgor, dry skin, and dry mucous membranes

Fluid volume deficit

Complications:

- Hypovolemic Shock (is a life-threatening emergency)
- Death



Fluid volume deficit

Nursing management

- Administer **oxygen**.
- Provide **fluid replacement** : oral if conscious and parenteral (whole blood, packed RBCs, plasma, ringer , lactate, normal saline)
- Administer **vasoconstrictors**, such as dopamine and norepinephrine; coronary vasodilators.



Fluid volume deficit

Nursing management

- Assess **skin** and **mucous membranes** for dryness
- Assess **capillary refill** appropriately.
- Monitor **intake** and **output**.
- Monitor serum **electrolytes**, BUN, creatinine, urine electrolytes
- Monitor **cardiac rhythm** if there is an electrolyte disturbance.

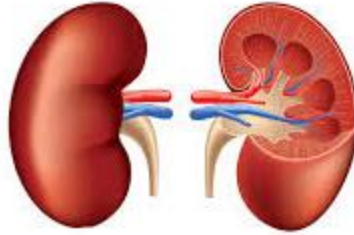


Fluid volume excesses

Excessive retention of water & sodium

Causes:

- Renal failure



- Excessive sodium intake.



- Age-related changes in cardiovascular and renal function.

Fluid volume excesses

Clinical manifestation:

- **Vital signs:** hypertension, tachypnea, tachycardia, Increased central venous pressure.
- **Neuromusculoskeletal:** confusion, muscle weakness, headache
- **Respiratory:** dyspnea, orthopnea, crackles
- **Edema, distended neck veins, pale and cool skin.**



Fluid volume excesses

Nursing management

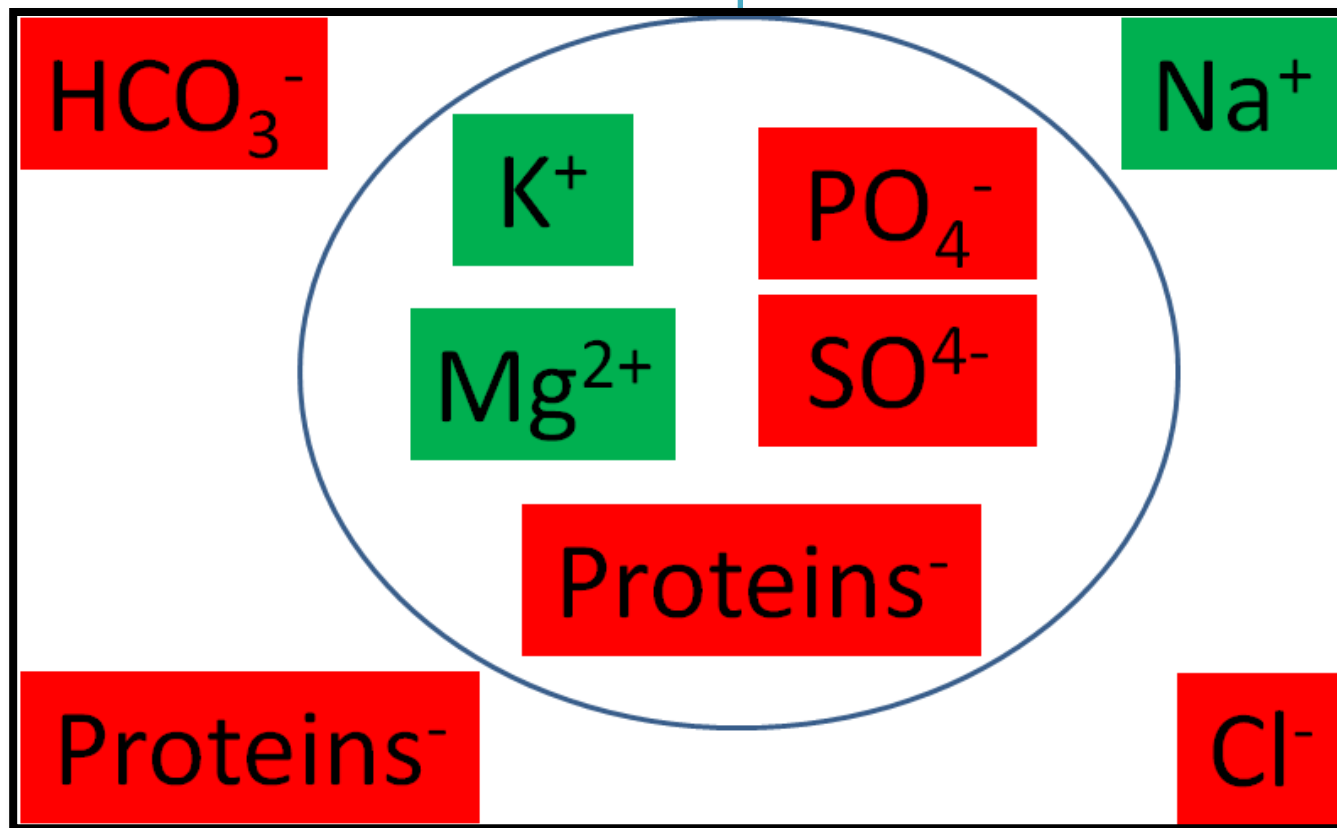
- Monitor **intake** and **output**.
- Administer **oxygen** as prescribed.
- **Reduce IV** flow rates.
- Administer **diuretics** (osmotic, loop) as prescribed.
- **Limit** fluid and sodium intake as prescribed.
- **Reposition** the client at least every 2 hr.
- Monitor **vital signs** and heart rhythm.
- Encourage clients to **weigh themselves daily**.

Intra cellular Electrolytes

- Potassium (K^+)
- Magnesium (Mg^{++})
- Phosphate (PO_4)

Extra cellular Electrolytes

- Calcium (Ca^{++})
- Sodium (Na^+)
- Bicarbonate (HCO_3)
- Chloride (Cl)



ELECTROLYTE IMBALANCE





The Roles of Electrolytes



Calcium

Contracting muscles, signaling nerves, blood clotting, forming & maintaining bones & teeth

Sodium

Maintain fluid balance, support muscle contractions, helps signal nerves

Potassium

Helps maintain blood pressure, regulates heart contractions, assists with muscle functions

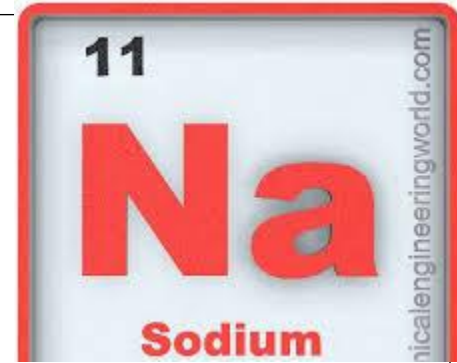
Magnesium

Support muscle contractions, maintain proper heart rhythms, supports nerve functions, helps fluid balance

Sodium Imbalance

Hyponatremia

Hypernatremia



Hyponatremia occurs when the sodium level is **less than 135 mEq/L**.

Causes:

1. **GIT loss:** as vomiting, diarrhea
2. **Na loss in urine:** with renal failure, diuretics
3. **Blood loss.**
4. **Hypovolemia & excessive diaphoresis.**
5. **Restricted dietary intake** (Restricted Na^+ in diet).

Clinical manifestation of hyponatremia

- **Vital signs:** hypotension, tachypnea, tachycardia, hypovolemia.
- **Oliguria**
- **Dry skin & lax** S.C tissue.
- **Weakness**, nausea, vomiting, and diarrhea may also occur due to increased gastrointestinal motility.
- **Sunken eyes**



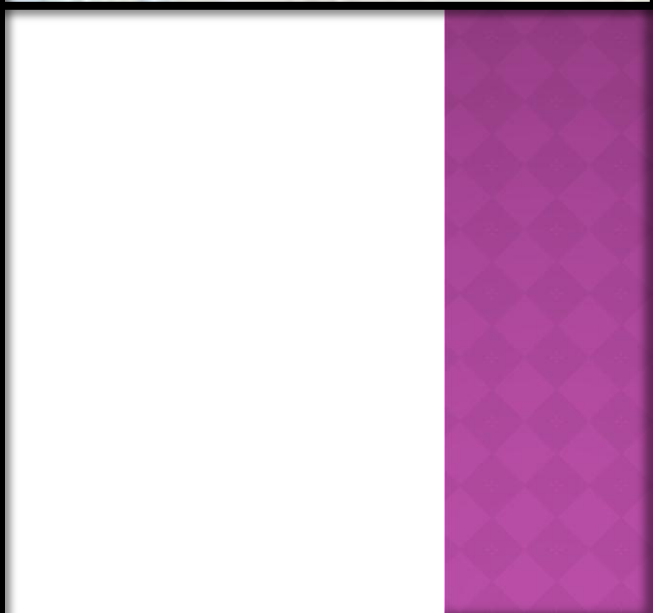
Hypernatremia occurs when the sodium level is **above 145 mEq/L**.

Causes:

- **Excessive saline** transfusion in the early post-operative period
- Na^+ **resorption** with (Aldosterone or cortisone)
- Inability to **excrete Na^+** .
- Abnormal renal **retention** of Na^+ .

Clinical manifestation of hypernatremia

- Slight **puffiness** of the face
- **Edema**
- **Ascites.**
- Weight **gain** due to fluid retention
- Hypertension
- **Mental** status **changes**, such as agitation, confusion, and personality changes.



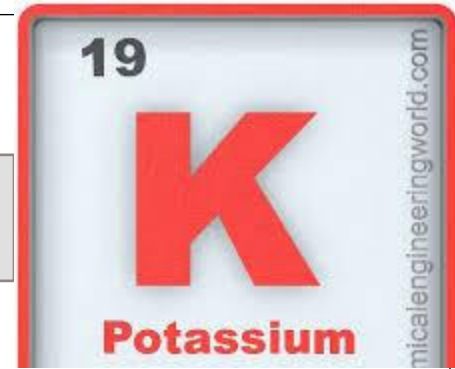
Nursing management

- Report abnormal **laboratory** findings to the provider.
- Monitor level of **consciousness** and ensure safety.
- Monitor **vital signs** and heart rhythm.
- Provide **oral hygiene** and other comfort measures to decrease thirst.
- Monitor **intake and output**
- **Weigh** daily and compare

Potassium Imbalance

Hypokalemia

Hyperkalemia



Hypokalemia

occurs when the serum potassium level below 3.5 meq/l

Causes:

1. **GIT loss:** as vomiting, diarrhea.
2. **Certain types of medications (lasix)**
3. **Major surgery and hemorrhage.**
4. **Excessive diaphoresis.**
5. **Restricted dietary intake (rare).**

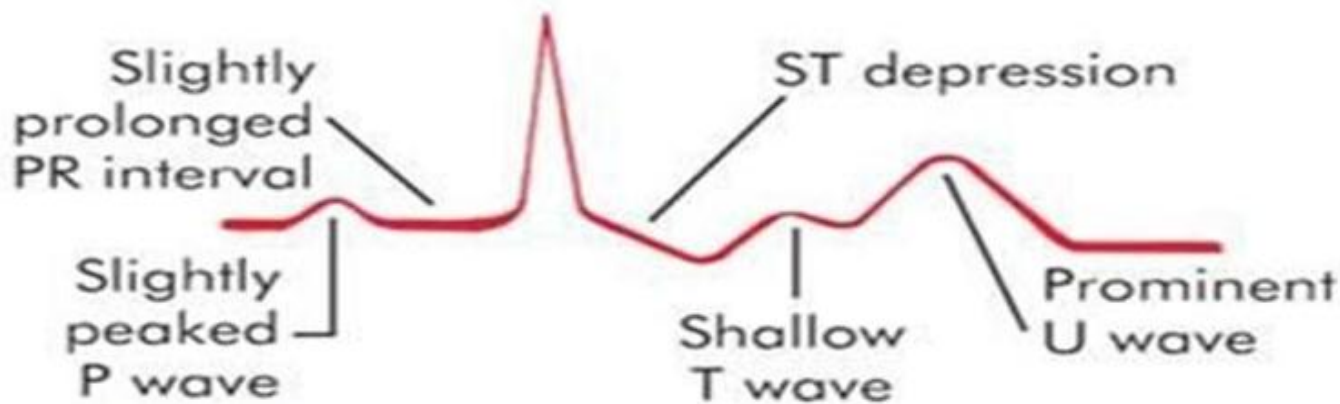
Clinical manifestation of hypernatremia

- Most patients are **asymptomatic**.
- **Malaise** and **weakness**.
- The motility of the GI system is **slowed**, causing nausea, vomiting, abdominal distention, and constipation.
- **Muscle cramping**

Irregular heartbeat (**dysrhythmia**), which can lead to a **cardiac arrest**

ECG changes: depression of the ST segment, and inversion of the T wave.

Hypokalemia



⦿ Medical treatment:

- ✓ Oral potassium supplements are given.
- ✓ Correct fluid imbalance.
- ✓ Monitor cardiac rhythm.
- ✓ Stop or change medications that contribute to potassium loss, if possible (as diuretics).
- ✓ Encourage potassium-rich foods.
- ✓ Administer potassium supplements.
- ✓ Administer potassium in intravenous fluid



Potassium is **never** given by IV push : because; Hyperkalemia may cause cardiac **arrhythmias** .Potassium is very **irritant** to the vein causing **thrombophlebitis**.



⊙ Precautions for patients receiving oral potassium

- ✓ Given in **juice** or other desired liquids to improve taste and to prevent gastrointestinal irritation.
- ✓ Take it with **meals**.
- ✓ Report **adverse effects**, such as nausea, vomiting, diarrhea, and abdominal cramping

Nursing management

- Monitor cardiac system for rate, rhythm
- Monitor **vital signs** and heart rhythm.
- Monitor **intake and output**
- Monitor intravenous site for redness, swelling, warmth, and pain.
- Monitor level of **consciousness**
- Provide oral potassium supplementation.

Foods **rich** in potassium (bananas, tomatoes, orange juice ,avocados, broccoli, cantaloupe.)



Hyperkalemia

occurs when the serum potassium level above **5 meq/l**

Causes:

1. **Overuse of potassium.**
2. **Excessive intake** of oral or IV potassium supplements
3. **Patients with renal failure** are at risk for hyperkalemia because the kidneys cannot excrete potassium.

Clinical manifestation of hyperkalemia

- Muscle **twitches** and **cramps**, followed by muscular **weakness**.
- Increased GI motility; **diarrhea**, **Nausea**, **vomiting**.
- **Abdominal distention**.
- Slow, **irregular** heart rate, and decreased blood pressure.
- **Weakness** and **dizziness**

ECG changes:

High peaked T wave.

Wide QRS complex.

Depressed ST segment

Flat P wave.

Hyperkalemia



◎ Medical treatment:

- ✓ I. V. **calcium gluconate** : Antagonizes effect of potassium on heart.
- ✓ IV **sodium bicarbonate** “ NaHCO_3 ”: shifts potassium intracellular.
- ✓ **Diuretics**
- ✓ Dietary **limitation** of potassium-rich foods
- ✓ **Dialysis**

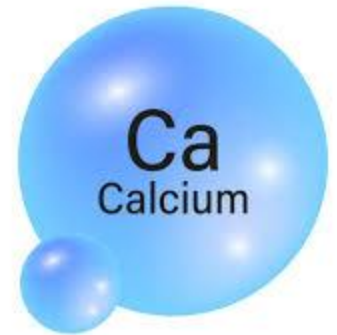
Nursing management

- Monitor **vital signs**, cardiac rhythm
- Monitor abdominal **bowel sounds**, distention, and pain.
- Monitor **intravenous site** for redness, swelling, and pain.
- **Decrease** potassium intake.
- **Stop** the infusion of IV potassium.
- **Withhold** oral potassium.
- Provide a **potassium-restricted diet**

Calcium Imbalance

Hypocalcemia

Hypercalcemia



Hypocalcemia

occurs when the serum calcium level below 4.5 meq/l

Causes:

- Inadequate intake of calcium.
- Post thyroidectomy.
- Vitamin D deficiency: Prevents calcium absorption.
- The woman who is postmenopausal
- Inadequate absorption of calcium from the intestine.

Hypocalcemia Signs & Symptoms "CATS"



Convulsions

Arrhythmias

Tetany

Spasms

Chvostek's sign

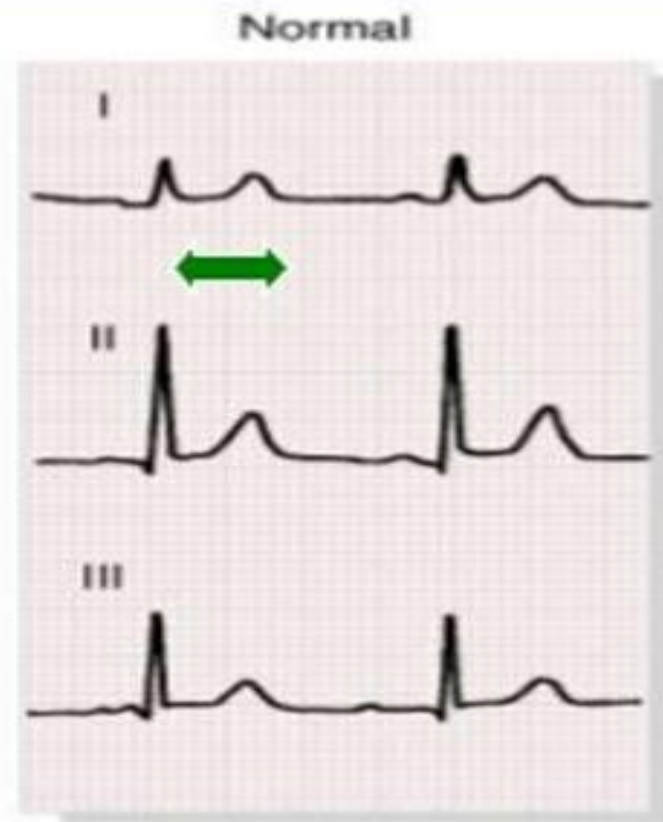
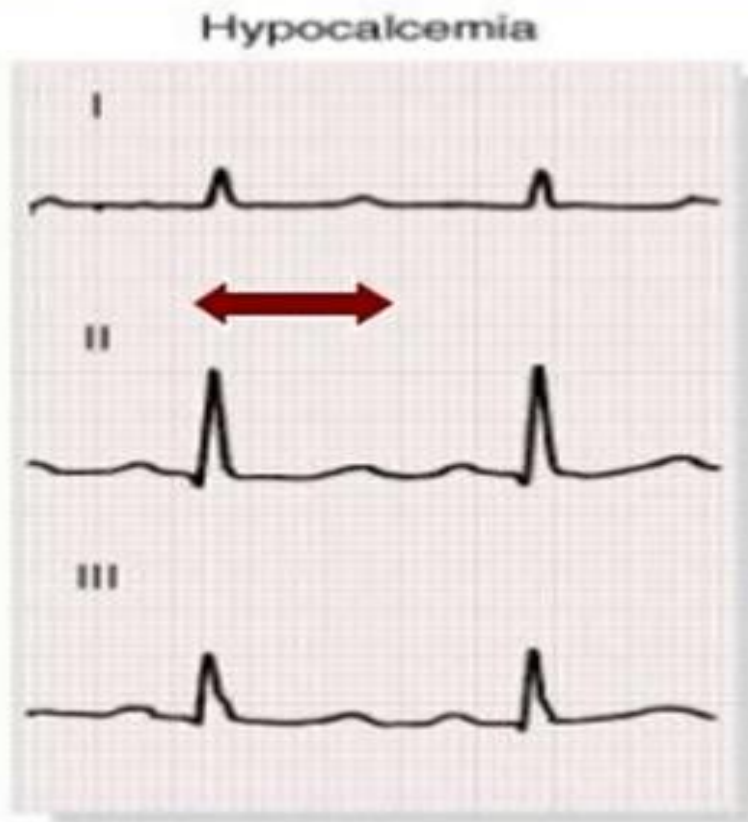


An indication of tetany in which a unilateral spasm of the oris muscle is initiated by a slight tap over the facial nerve anterior to the external auditory canal.

- **Trousseau's sign** consists of carpal spasm provoked by ischemia (induced by inflation of the blood pressure measuring cuff around the arm) or alkalosis (provoked by hyperventilation for a few minutes)



ECG: shows prolonged QT interval, Prolonged ST interval.



- **Medical treatment:**

- ✓ **Oral calcium** supplements should be administered
1 : 2 hours after meals or 30 min before meals
- ✓ High calcium **diet**
- ✓ **Vitamin D** foods are also encouraged, especially milk and other dairy products. helps with absorption of calcium
- ✓ IV injection of calcium gluconate or calcium chloride.

Hypercalcemia

occurs when the serum calcium level above 5.5 meq/l

Causes:

- Excessive intake of calcium or vitamin D
- Hyperparathyroidism
- In patients with invasive or metastatic cancer.
- Overuse or prolonged use of diuretics



Drowsiness or confusion



Palpitation or irregular heartbeat



Loss of concentration or interest



Muscle spasms



Thirsty

SYMPTOMS OF HYPERCALCEMIA



Loss of appetite



Polyuria



Low mood or increase in irritability



Constipation



Nausea and vomiting

Medical treatment:

- ⊙ **Stop** all calcium-containing medications (supplements, antacids).
- ⊙ **Monitor** cardiac rhythm.
- ⊙ Administer **normal saline** as it aids in urinary excretion of calcium.
- ⊙ Administer **diuretics** to enhance the excretion of calcium
- ⊙ Administer calcitonin, phosphorus, bisphosphonates to inhibit calcium resorption from the bone .

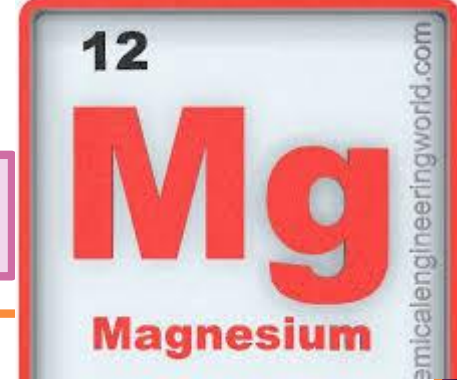
- Severe hypercalcemia: **dialysis**, may be necessary to cleanse the blood of excess calcium



Magnesium Imbalance

Hypomagnesemia

Hypermagnesemia



Hypomagnesemia

occurs when the serum magnesium below 1.7 meq/l

Causes:

- **Decreased** intake, including malnutrition and starvation diets.
- Excessive **loss** of magnesium.
- Severe **diarrhea**
- **Alcoholism**, which causes increase renal excretion of magnesium.
- Certain **drugs**, such as diuretics, and some anticancer agents ;increase renal excretion of magnesium

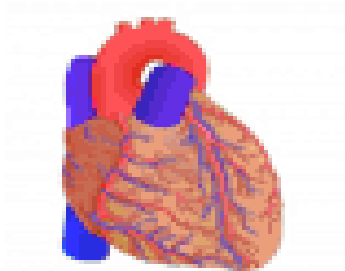
Hypomagnesemia

↪ Serum level < 1.7 mg/dL



Neuromuscular manifestations

- Tremor, tetany, seizures
- Weakness
- Apathy
- Delirium
- Coma



Cardiovascular manifestations

- Prolonged QTc
- Widening of QRS
- Atrial and ventricular dysrhythmias

Nursing management

- Monitor **vital signs**, cardiac rhythm
- Monitor **intake & output**
- Discontinue magnesium-losing medications (e.g., diuretics).
- Administer oral or IV magnesium
- **IV route is used because IM can cause pain and tissue damage.**
- Encourage foods high in magnesium.



yogurt



pumpkin seeds



lentils



bananas



whole grains

food
sources
of
Magnesium
(Mg)



dark chocolate



spinach



figs



almonds



black beans



avocados



fish

Hypermagnesemia

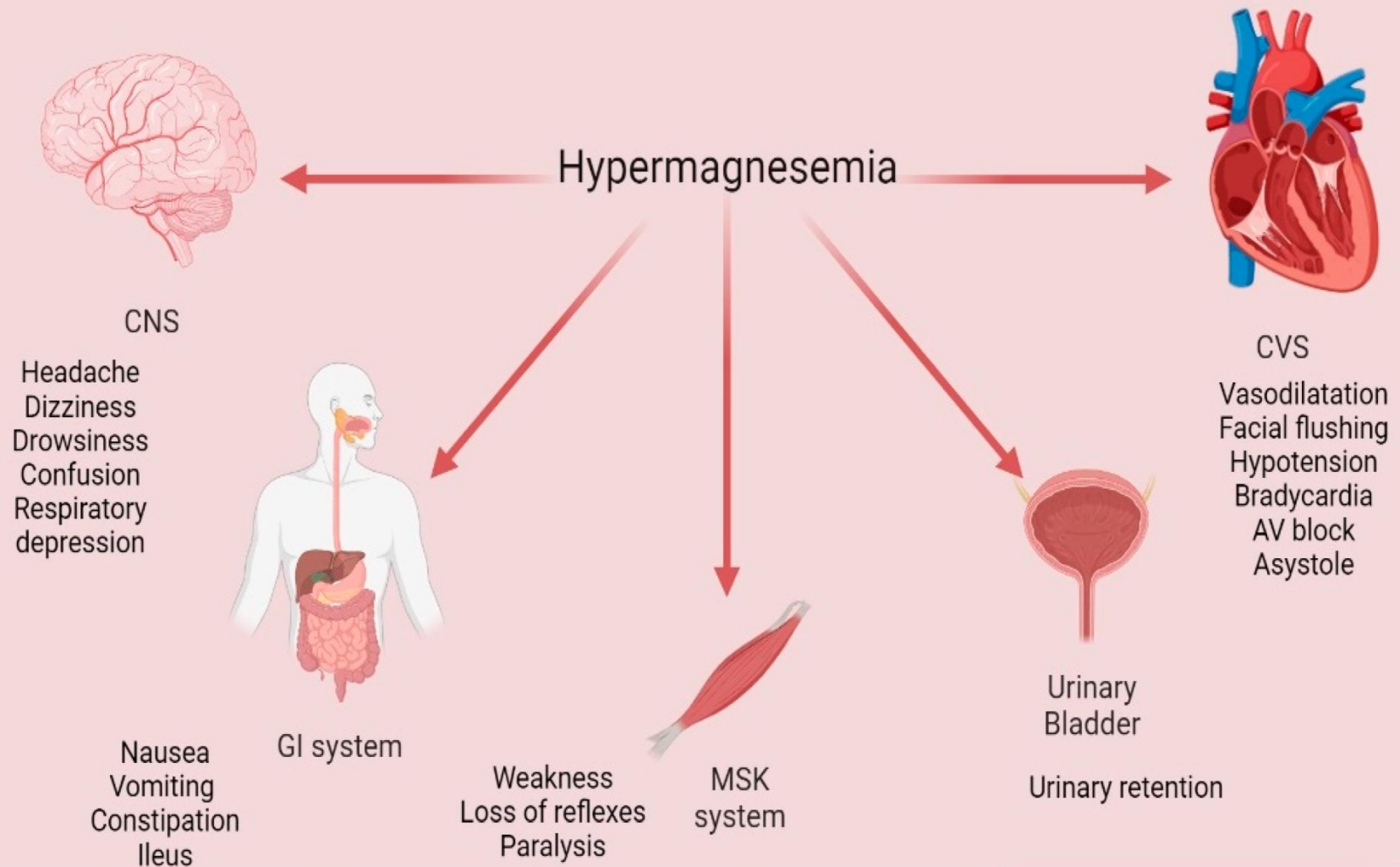
occurs when the serum magnesium above **2.5 meq/l**

Causes:

- Increased intake
- Decreased renal excretion caused by renal failure

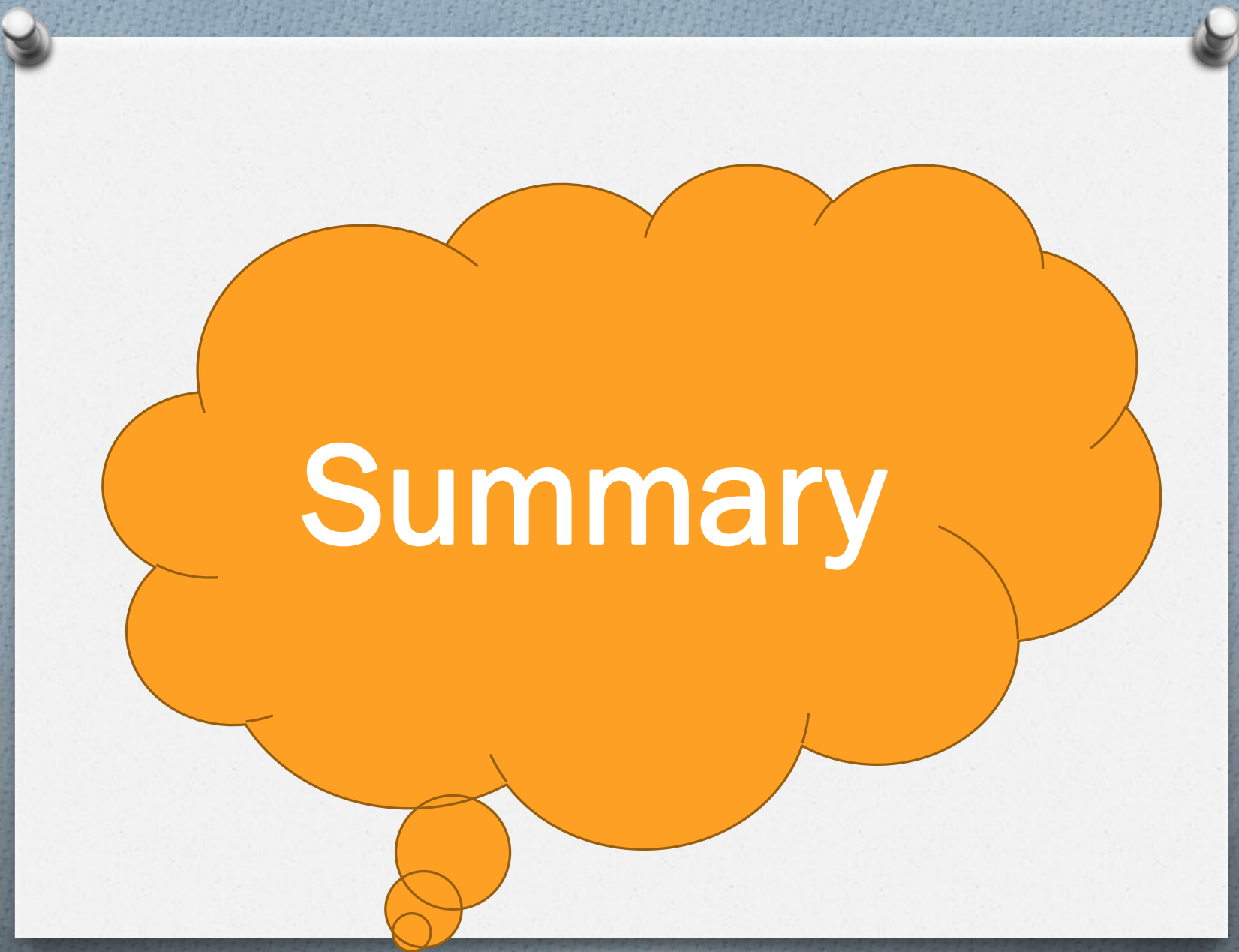


Clinical Manifestations of hypermagnesemia

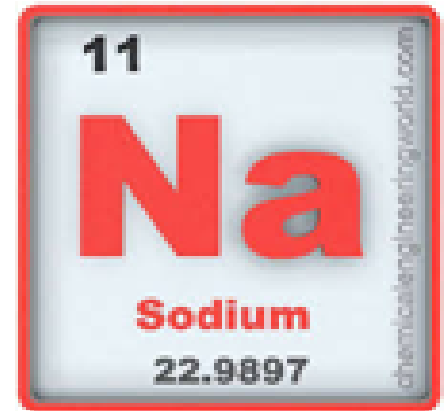


Nursing management

- Monitor vital signs, cardiac rhythm
- Monitor intake & output
- Diuretics.
- Dialysis
- Avoid foods high in magnesium.



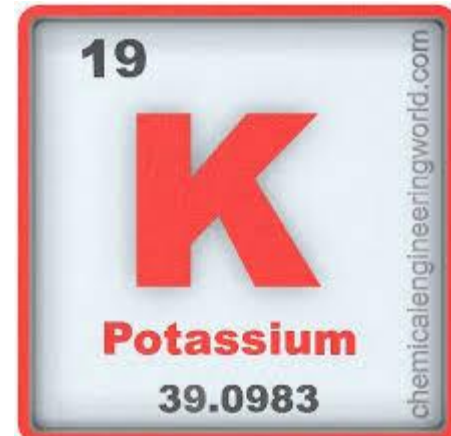
Summary



Sodium Imbalance

Hyponatremia

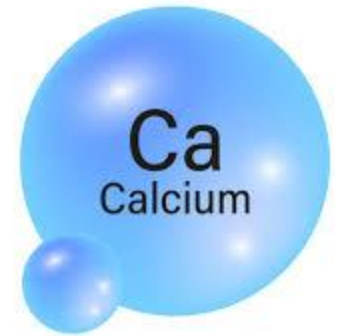
Hypernatremia



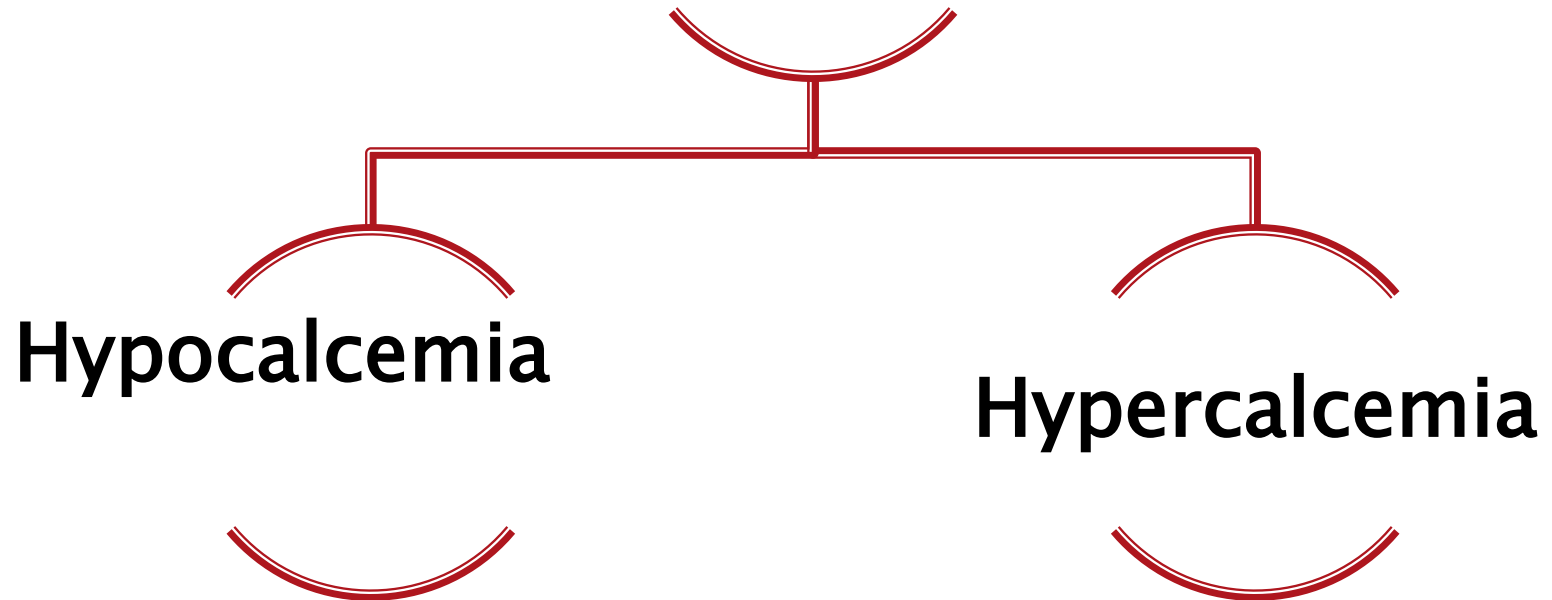
Potassium Imbalance

Hypokalemia

Hyperkalemia



Calcium Imbalance





Magnesium Imbalance

Hypomagnesemia

Hypermagnesemia



thank
you